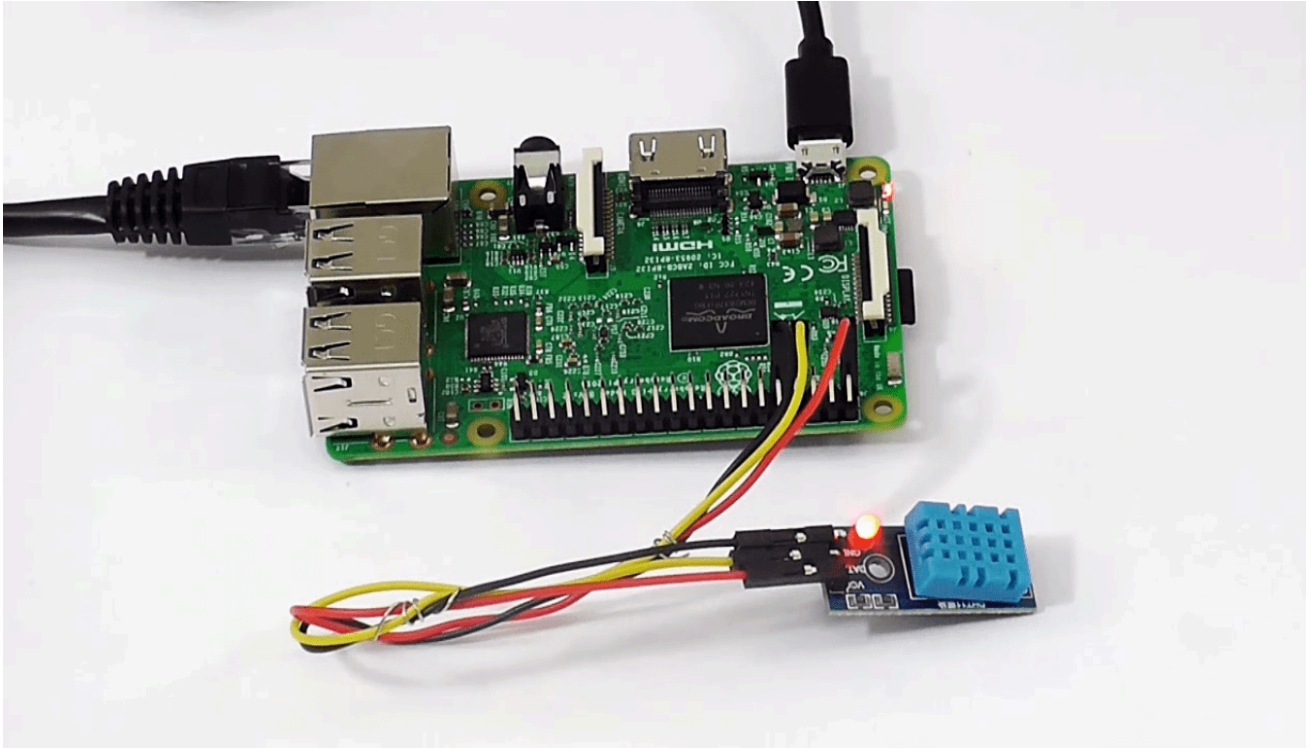


CampusConnect



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1. Description of the IoT system and its purpose.

CampusConnect transforms the University of Deusto campus into an adaptive, human-centered network of smart spaces aimed at strengthening social cohesion. The system continuously senses environmental and occupancy-related variables in shared locations (for example: library areas, cafeterias, the Plaza de la Memoria/clostros and study rooms) and presents the resulting information through a public dashboard and local visual cues. By making the communal state of each area visible—whether quiet, active, crowded or ideal for socializing—the project helps students and staff make informed decisions about where to study, relax or meet peers, thereby improving usage of common spaces and increasing spontaneous social interactions.

At a technical level, CampusConnect integrates a distributed network of sensors that measure noise level, occupancy, temperature, light, and air quality, together with simple actuators (LED indicators and screens) that provide intuitive, local feedback. Sensor nodes publish aggregated readings via Wi-Fi using MQTT to a backend ingestion service which stores time-series data for live visualization and historical analysis (Grafana).

To validate the concept within the scope of this course, a functional prototype (Minimum Viable Product – MVP) will be developed.

The MVP will focus on two representative campus areas equipped with a limited set of sensors and actuators to demonstrate the system's core capabilities. Specifically, the prototype will include real-time data collection (noise, occupancy, air quality, and temperature), transmission through a Wi-Fi connection, local data storage and visualization through a public dashboard. Local actuation will be provided by LED indicators showing the current state of each monitored area using a simple green/yellow/red color logic. This proof of concept will serve as a small-scale but realistic demonstration of how IoT can make the campus environment more transparent, inclusive and collaborative.

Beyond environmental monitoring, CampusConnect is explicitly social: it promotes positive behaviors by recognizing collective good practices and it emphasizes privacy by aggregating and anonymizing all data. Unlike traditional environmental control systems that optimize for comfort alone, CampusConnect centers social interaction and community stewardship—using IoT not only to measure the campus environment but to actively foster a more connected, welcoming, and collaborative campus culture.

2. Use case or real-life scenario.

To illustrate the potential impact of CampusConnect, the following scenario represents a typical day on campus and how different members of the community interact with the system.

Scenario 1 – The Student Searching for Focus

Alfreda, a third-year computer engineering student, arrives at the University of Deusto in the afternoon to study between classes. She opens the dashboard displayed on a nearby screen and checks which areas are quiet. The dashboard shows that the main library is moderately busy, while one of the study rooms in the Claustros is calm and well-ventilated. Alfreda chooses to go there, finding an ideal environment to concentrate.

While she studies, the noise sensor detects that the area remains quiet and the LED bar installed in the room glows green, signaling to others that it is a peaceful space. This helps maintain the atmosphere without any verbal reminders.

Scenario 2 – The Group Project Meeting

Later that day, a group of students from the architecture faculty gathers in a shared work area to discuss their group project. As they begin collaborating, the system detects an increase in sound and occupancy, shifting the LED indicator from green to yellow. Seeing this, the group decides to lower their voices.

At the end of the session, the system records that the area stayed within acceptable noise levels for most of the time. On a nearby display, a brief message appears: *“Thank you for keeping this space collaborative and comfortable for everyone!”* This instant feedback reinforces positive group behavior and respect for shared spaces.

Scenario 3 – Facilities and Maintenance Staff

Meanwhile, the campus maintenance staff use the same dashboard to monitor air quality and temperature across different areas. When the gas and temperature sensors in the cafeteria detect poor ventilation and slightly high temperature, an automatic alert appears on their internal dashboard. They decide to open windows and adjust the ventilation system.

This not only improves comfort but also optimizes resource management and encourages a shared responsibility for the wellbeing of the university community.

Through these interactions, CampusConnect enables all members of the university—students, faculty, and staff—to make informed and collaborative decisions. The system fosters awareness, encourages small daily actions that strengthen trust, and builds a sense of belonging on campus. It transforms the physical spaces of the university into living environments that react to people's needs and behaviors, connecting technology and community in a meaningful way.

3. Advantages compared to existing systems.

CampusConnect stands out from other systems because it focuses not only on monitoring the environment, but also on improving social interaction and community life. Most existing smart campus or building systems are designed to control temperature, lighting, or energy use. They make spaces more efficient, but they rarely help people connect or collaborate.

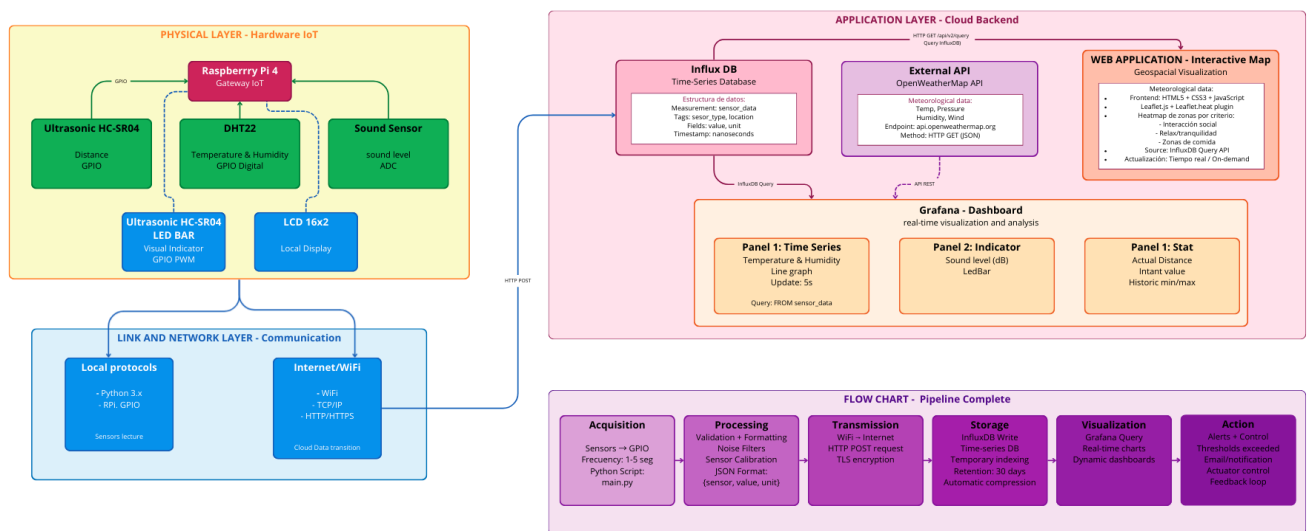
Our system goes a step further by using environmental data to encourage positive human behavior. For example, instead of simply showing a noise level, CampusConnect gives visual feedback with colored LEDs or thank-you messages that make users aware of how their actions affect others. This transforms the sensors from passive devices into active tools for social awareness.

Another advantage is that the system uses simple and affordable technology. It can be easily installed in different areas of the university and adapted to new situations. The use of Wi-Fi and

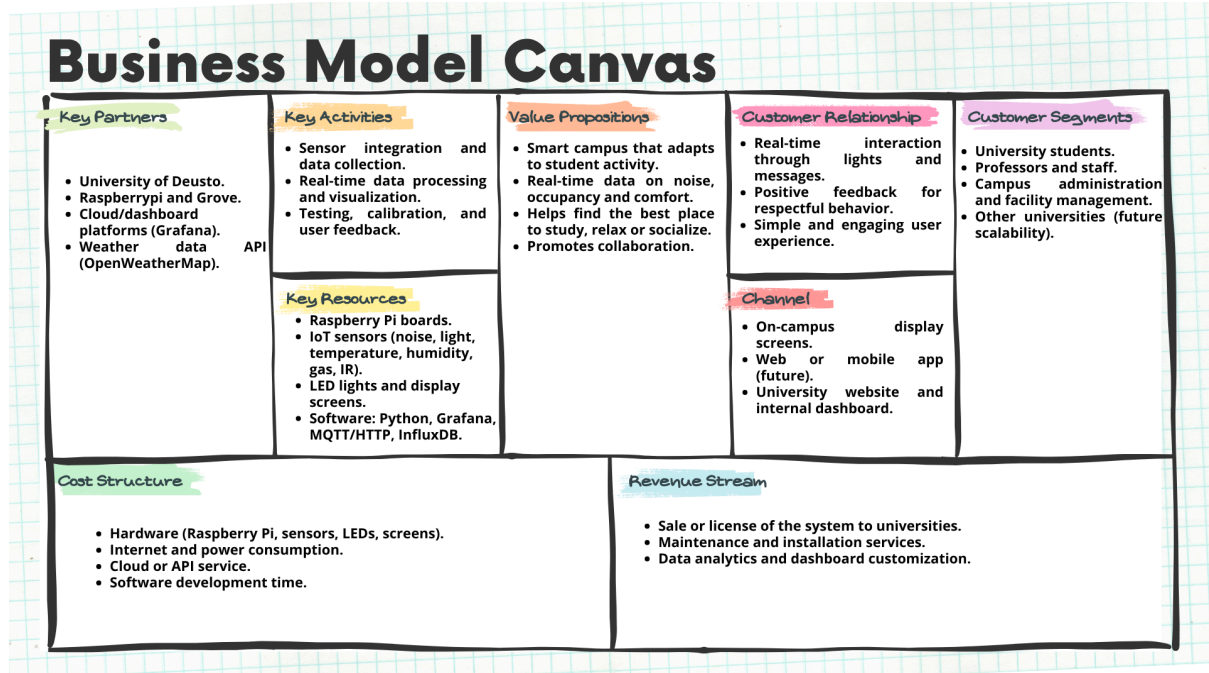
MQTT makes it flexible and scalable, while the dashboard (Grafana or ThingsBoard) allows everyone to see the same information in real time.

Finally, CampusConnect respects privacy because it works with anonymous and aggregated data. The goal is not to monitor individuals, but to improve shared experiences. This combination of low-cost IoT technology, positive feedback, and focus on community wellbeing makes CampusConnect different from traditional environmental monitoring systems.

4. Solution architecture diagram.



5. Business model (Canvas).



6. Initial task planning (SCRUM + MVPs).

Initial task planning following SCRUM methodology and MVPs

Sprint 1:

- **Sprint duration:** 5 nov, 2026 - 19 nov, 2025
- **Focus:** Initial configurations, basic setup, and test script creation.

User Story 1: Initial configuration of the project

Objective: Set up the initial project structure, environment, and configurations.

- **Tasks:**
 1. **Create folder structure** – [Issue](#)
 2. **Create Python project** – [Issue](#)
 3. **Configure the GitHub project** – [Issue](#)

User Story 2: Documentation

Objective: Prepare and update project documentation for future development and deployment.

- **Tasks:**
 1. **Make the Business Model** – [Issue](#)
 2. **Make the Project Documentation** – [Issue](#)
 3. **Make the architecture Diagram**– [Issue](#)
 4. **Make the Heat Map**-

User Story 3: Initial Testing Setup

Objective: Develop core scripts for sensor functionality and define the project's architecture.

- **Tasks:**
 1. **Create Led Bar Initial Test Script** – [Issue](#)
 2. **Create Noise Initial Tets Script** – [Issue](#)
 3. **Create Air Quality Initial Test Script** – [Issue](#)
 4. **Create Temperature and Humidity Initial Test Script** – [Issue](#)
 5. **Create Display Test Script** – [Issue](#)

Sprint 2:

- **Sprint duration:** 5 nov, 2026 - 19 nov, 2025
- **Focus:** Initial configurations, basic setup, and test script creation.

User Story 3: Configure the classes

Objective: Create initial test scripts for various sensors and hardware components.

- **Tasks:**
 1. **Create Main Script-** [Issue](#)
 2. **Create Noise Script –** [Issue](#)
 3. **Create Led Bar Script –** [Issue](#)
 4. **Create Air Quality Script –** [Issue](#)
 5. **Create Temperature and Humidity Script –** [Issue](#)
 6. **Create Display Script –** [Issue](#)
 7. **Integrate external API (Weather) with the application-** [Issue](#)
 8. **Visualize metrics in Grafana-** [Issue](#)

Sprint 3:

- **Sprint duration:** 5 nov, 2026 - 19 nov, 2025
- **Focus:** Initial configurations, basic setup, and test script creation.

User Story 5: Create Database, and Visualization

Objective: Set up the database, integrate external APIs, and implement data visualization features.

- **Tasks:**
 1. **Create the Database script –** [Issue](#)
 2. **Use InfluxDB as the Database –** [Issue](#)
 3. **Create the Database Table –** [Issue](#)

Dependencies Overview

The project is organized using SCRUM, and the work is divided into several user stories with clear dependencies:

User Story 1 – Project Setup

This is the foundation of the whole project. All tasks in User Story 1 must be completed before starting any other user story.

It includes creating the folder structure, configuring the GitHub project, and setting up the

Python environment.

These tasks ensure that the team has a stable and organized base to work from.

User Story 2 – Documentation

Documentation runs in parallel to the other user stories.

Although it does not block development directly, it depends on progress from the setup, testing, and implementation tasks.

As more features are completed, the documentation is updated.

User Story 3 – Test Scripts (Initial Hardware Testing)

This user story depends fully on User Story 1.

Once the project structure and environment are ready, the team can develop and run test scripts for the sensors.

Each test script (noise sensor, LED bar, air quality sensor, temperature and humidity sensor, and the LCD display) depends on having the physical sensor correctly connected and configured.

These tests help verify that all components work before building more advanced features.

User Story 4 – Sensor Scripts and System Architecture

User Story 4 depends on both User Story 1 (setup) and User Story 3 (test scripts).

The results of the test scripts guide how the final sensor scripts are written.

This user story also includes designing the system architecture and structuring the classes needed for the main application.

All this must be consistent with the hardware tests already done.

User Story 5 – Database, API Integration, and Visualization

This user story requires that User Story 1 is completed to prepare the basic configuration of the database environment.

It also depends on User Story 3 and User Story 4, because the database needs real sensor data to be stored correctly.

In addition, API integration (like the weather API) can only be done once the data flow from the sensors is set up.

Finally, visualization in Grafana depends on both the database setup and the availability of stored sensor data.

Sprint Planning Summary

The project is organized into three MVP stages, each one completed in a sprint:

MVP 1 – Sprint 1: Project Setup, Documentation, and Initial Test Scripts

This MVP focuses on building the foundation of the project.

During Sprint 1, the team completed User Story 1 (project setup) and started working on

documentation and hardware test scripts.

Examples of completed tasks include creating the folder structure, configuring the GitHub project, and testing basic sensor functions (noise, LED bar, air quality, temperature/humidity, and display).

This MVP provides proof that the project environment is ready and that the sensors work correctly.

MVP 2 – Sprint 2: Sensor Scripts and Architecture Design

Sprint 2 focuses on development.

The team works on User Story 4, creating the full sensor scripts based on the previous test scripts.

They also create the system architecture diagram and begin writing the main script that will combine all sensor readings.

Some class configuration tasks also belong to this sprint.

By the end of MVP 2, the system can read sensors continuously, update their values, and prepare the data for storage.

MVP 3 – Sprint 3: Database, External API, and Visualization

In Sprint 3, the team completes User Story 5.

The tasks include creating the database, setting up the tables, writing the database script, and integrating the weather API into the application.

Finally, the team visualizes all metrics in Grafana.

At the end of this sprint, the project has a functional backend, real-time data flow, and a complete dashboard showing the final results.

7. References and links.

[1] "[Products] MY-SEMI INC.--Best LED Driver Through MY-Semi.," *My-semi.com*, 2025.
http://www.my-semi.com/content/products/product_mean.aspx?id=4 (accessed Dec. 8, 2025).

[2] "Grove - Barra LED | Seeed Studio Wiki," *Seeedstudio.com*, Jan. 07, 2023.
https://wiki.seeedstudio.com/es/Grove-LED_Bar/ (accessed Dec. 9, 2025).